

39. PHY Management Interface (PMGI)

This MCU has two PMGIs (PHY Management Interfaces) as interface circuits for access to registers in physical layer LSI (PHY LSI) chips for Ethernet links.

39.1 Overview

Table 39.1 lists the specifications of the PMGI, and Figure 39.1 is a block diagram of the PMGI.

Table 39.1 Specifications of the PMGI

Item	Description
Communications format	Supports the MII Management Interface compliant with IEEE 802.3.
I/O pins	MDC (clock output), and MDIO (data input and output)
Transfer rate	MDC is generated by dividing PCLKA. Division ratios are from 1/4 to 1/128.
PHY register address	5-bit address format
PHY device address	5-bit address format
Selection of inclusion or non-inclusion of a preamble	Inclusion or non-inclusion of a preamble is selectable.
MDIO output delay function	Delaying the timing of output on the MDIO pins is adjustable.
Adjustable timing for capture of the MDIO input	The timing for capture of the MDIO input is adjustable.
Interrupt sources	Completion of the sending of data for writing and of the reception of data being read.

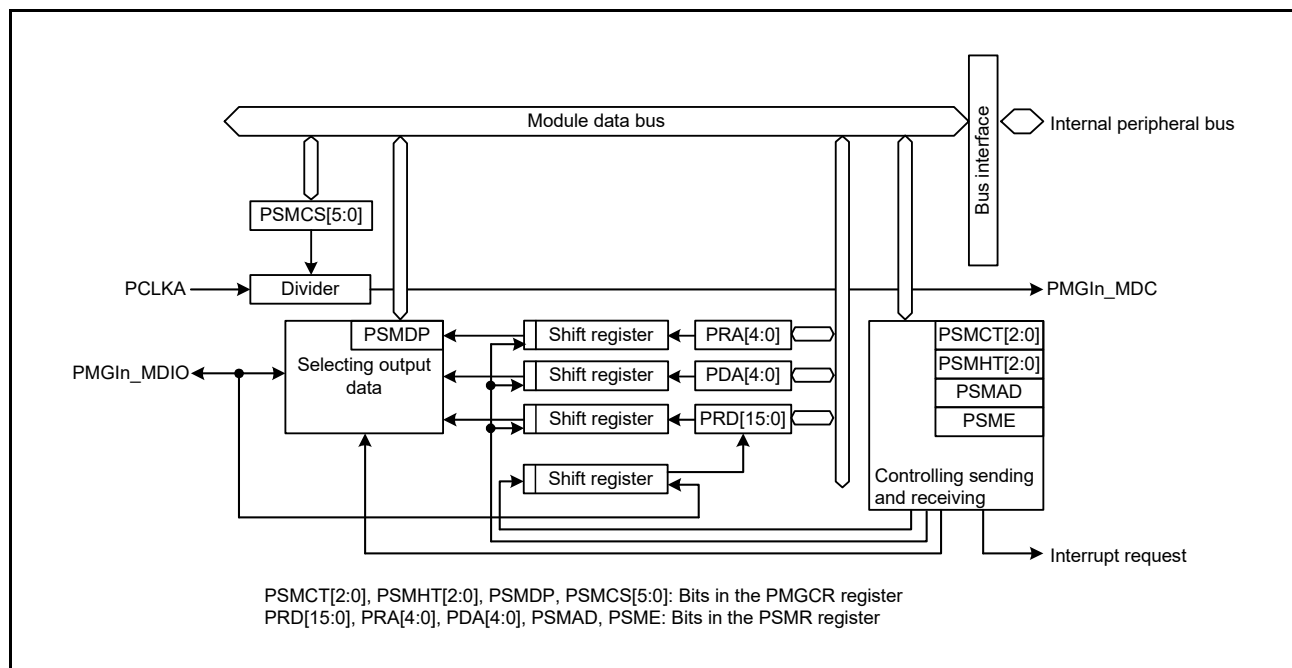


Figure 39.1 **Block Diagram of the PMGI ($n = 0, 1$)**

Table 39.2 lists the I/O pins of the PMGI.

Table 39.2 I/O pins of the PMGI

Channel	Pin Name	I/O	Description
PMGI0	PMGI0_MDC	Output	Output pin for the management data clock of PMGI0
	PMGI0_MDIO	I/O	I/O pin for the management data of PMGI0
PMGI1	PMGI1_MDC	Output	Output pin for the management data clock of PMGI1
	PMGI1_MDIO	I/O	I/O pin for the management data of PMGI1

39.2 Register Descriptions

39.2.1 PMGI Configuration Register (PMGCR)

Address(es): PMGI0.PMGCR 000C 5880h, PMGI1.PMGCR 000C 5890h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	PSMCT[2:0]		—	PSMHT[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	PSMDP	—	PSMCS[5:0]					—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13 to b8	PSMCS[5:0]	PHY Station Management Clock Select	The clock frequency of the MDC (clock output) is selected. Taking the value of the PSMCS[5:0] bits as n: if n is 000000b, the clock signal is fixed at low-level. Values of n other than 000000b select a clock signal with the frequency $PCLKA / ((n + 1) \times 2)$ as the MDC (clock output).	R/W
b14	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b15	PSMDP	PHY Station Management Preamble Control	0: Preamble field included 1: Preamble field not included	R/W
b18 to b16	PSMHT[2:0]	PHY Station Management Hold Time Adjustment	b18 b16 0 0 0: No output delay 0 0 1: 1 cycle of PCLKA 0 1 0: 2 cycles of PCLKA 0 1 1: 3 cycles of PCLKA 1 0 0: 4 cycles of PCLKA 1 0 1: 5 cycles of PCLKA 1 1 0: 6 cycles of PCLKA 1 1 1: 7 cycles of PCLKA	R/W*1
b19	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b22 to b20	PSMCT[2:0]	PHY Station Management Capture Timing Adjustment	b22 b20 0 0 0: No adjustment 0 0 1: 1 cycle of PCLKA 0 1 0: 2 cycles of PCLKA 0 1 1: 3 cycles of PCLKA 1 0 0: 4 cycles of PCLKA 1 0 1: 5 cycles of PCLKA 1 1 0: 6 cycles of PCLKA 1 1 1: 7 cycles of PCLKA	R/W*2
b31 to b23	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Do not set the PSMHT[2:0] bits to a value greater than that of the PSMCS[5:0] bits.

Note 2. Do not set the PSMCT[2:0] bits to a value greater than that of the PSMCS[5:0] bits.

PSMCS[5:0] Bits (PHY Station Management Clock Select)

The PSMCS[5:0] bits select the frequency of the MDC (clock output). The MDC is generated by frequency-dividing PCLKA according to the setting of the PSMCS[5:0] bits. Refer to section 39.3.3, Setting the Management Data Clock (MDC), for example of settings and frequencies of the MDC.

PSMDP Bit (PHY Station Management Preamble Control)

The PSMDP bit selects the inclusion or non-inclusion of a preamble field.

PSMHT[2:0] Bits (PHY Station Management Hold Time Adjustment)

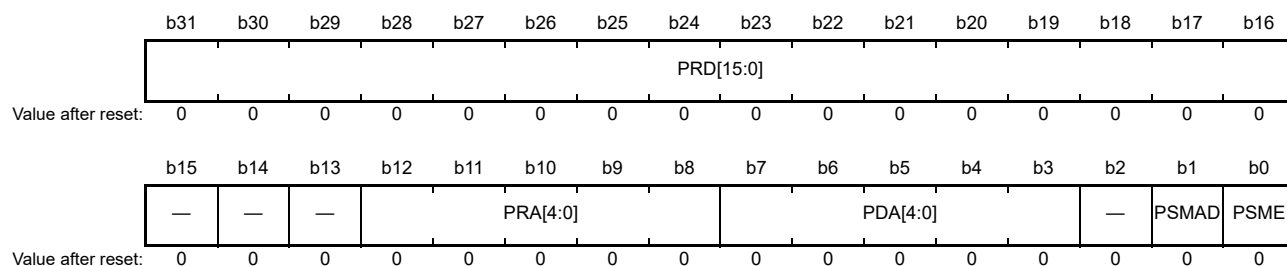
The output of the MDIO PSMHT[2:0] bits can be delayed by an amount controlled by the setting of the PSMHT[2:0] bits. These bits select the hold time for the output of the MDIO relative to rising edges of the MDC.

PSMCT[2:0] Bits (PHY Station Management Capture Timing Adjustment)

The PSMCT[2:0] bits select the timing for capture of the input on MDIO. The input on MDIO is captured before the number of PCLKA cycles from a rising edge of MDC.

39.2.2 PHY Station Management Register (PSMR)

Address(es): PMGI0.PSMR 000C 5884h, PMGI1.PSMR 000C 5894h



Bit	Symbol	Bit Name	Description	R/W
b0	PSME	PHY Station Management Start	0: Frame transfer for PHY station management completed 1: Start frame transfer for PHY station management	R/W
b1	PSMAD	PHY Station Management Access Direction	0: Reading 1: Writing	R/W
b2	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7 to b3	PDA[4:0]	PHY Address	Set the address of the PHY module	R/W
b12 to b8	PRA[4:0]	PHY Register Address	Set the address of the PHY register	R/W
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b31 to b16	PRD[15:0]	PHY Register Data	For reading: These bits hold the value read from a PHY register For writing: Set the value for writing to a PHY register in these bits	R/W

PSME Bit (PHY Station Management Start)

Setting the PSME bit to 1 triggers the start of the transfer of a frame for PHY station management. After the completion of transfer, this bit is automatically cleared to 0. Writing 0 to this bit is not possible. Only writing 1 to this bit is effective.

PSMAD Bit (PHY Station Management Access Direction)

The PSMAD bit selects the operation as reading or writing.

PDA[4:0] Bits (PHY Address)

The PDA[4:0] bits set the PHY address.

PRA[4:0] Bits (PHY Register Address)

The PRA[4:0] bits set the PHY register address.

PRD[15:0] Bits (PHY Register Data)

In the case of reading, the PRD[15:0] bits hold the value read from a PHY register after completion of the transfer of a frame for PHY station management. In the case of writing, these bits set the value for writing to a PHY register in the PRD[15:0] bits.

39.3 Operation

The PMGI supports MII management frame formats in compliance with IEEE 802.3, and handles access to the registers of PHY LSI chips via two serial interface circuits.

39.3.1 Format of MII Management Frames

Table 39.3 lists the data format of management frames, Figure 39.2 shows the format in the case of reading, and Figure 39.3 shows the format in the case of writing.

An MII management frame consists of a 32-bit preamble (PRE), 2-bit start field (ST), 2-bit operation code (OP) indicating reading or writing, 5-bit PHY address (PHYAD) and PHY register address (REGAD), 2-bit turnaround (TA) field, 16 bits of data (DATA), and a 1-bit idle condition (IDLE).

Setting the PSMR.PSME bit to 1 triggers the output of an MII management frame. The settings of the PSMR.PDA[4:0] bits and the PSMR.PRA[4:0] bits correspond to the PHY address and the PHY register address, respectively. In the case of reading, the value set in the PSMR.PRD[15:0] bits is written to the register of PHY-LSI. In the case of writing, the value to be written is set in the PSMR.PRD[15:0] bits.

The PMGCR.PSMDP bit can select if a preamble field is to be included. Setting the PMGCR.PSMDP bit to 1 triggers the start of sending from the start field.

Table 39.3 Data Format of Management Frames

Access Type	Management Frame								
	Item	PRE	ST	OP	PHYAD	REGAD	TA	DATA	IDLE
	Number of bits	32	2	2	5	5	2	16	1
Read		1..1	01	10	AAAAA	RRRRR	Z0	DDD..DDD	Z
Write		1..1	01	01	AAAAA	RRRRR	10	DDD..DDD	Z

PRE (Preamble): Send 32 consecutive 1's

ST (start of frame): Send 01b.

OP (operation code): Send 10b for reading or 01b for writing.

PHYAD (PHY address): These 5 bits select a PHY LSI chip.

REGAD (register address): These bits select one register from among up to 32 registers in the PHY LSI chip.

TA (turnaround): A 2-bit turnaround time to avoid contention between the register address and data during a read operation

(a) Send 10b when writing.

(b) Release the bus for 1 bit-period during a read operation (Z is output). This is indicated as Z0 because 0 is output by the PHY-LSI on the next clock cycle.

DATA (data): 16 bits of data. These bits are sequentially sent or received starting from the MSB.

IDLE (IDLE condition): Start of waiting time to input of the next MII management frame

(a) This releases the bus following writing (Z is output).

(b) The bus is released during TA before reading.

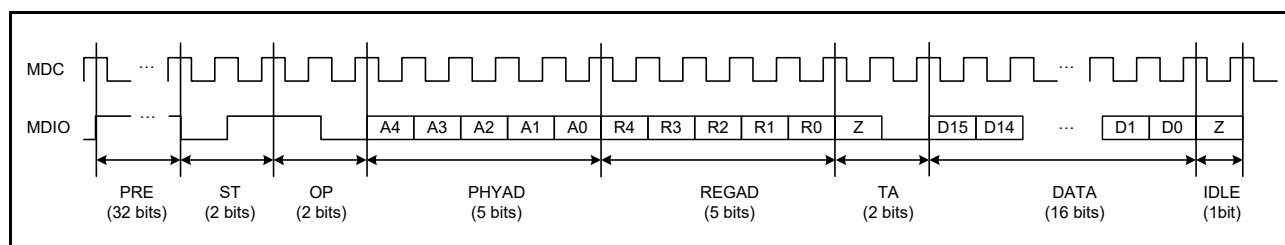


Figure 39.2 Data Format in the Case of Reading

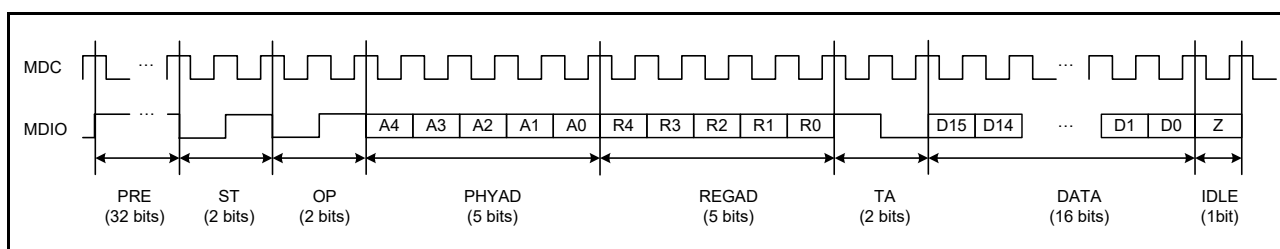


Figure 39.3 Data Format in the Case of Writing

39.3.2 PHY Register Access Procedures

39.3.2.1 Initializing a PMGI

Initialize a PMGI by following Figure 39.4, Flow Chart: Example of Initial Settings of a PMGI.

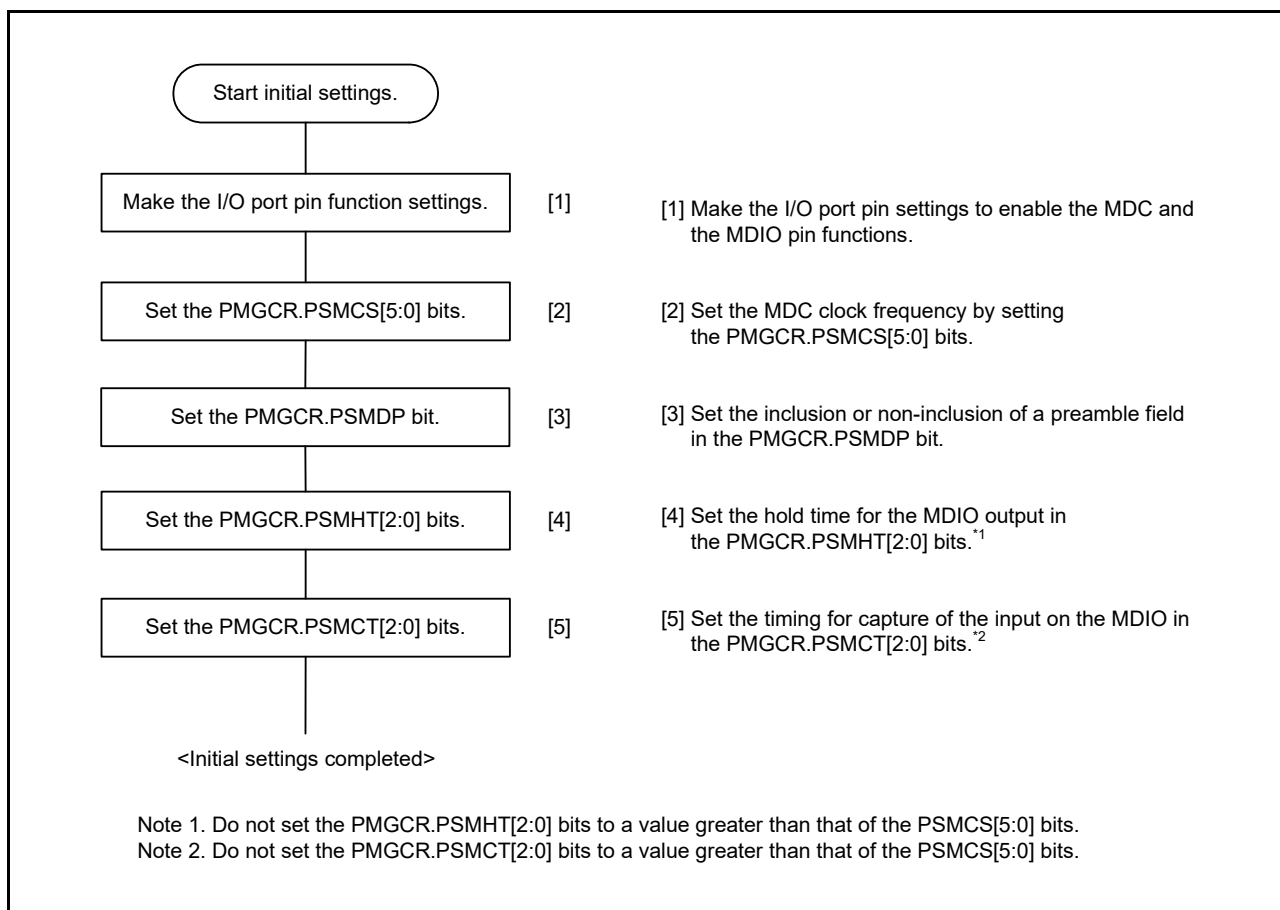


Figure 39.4 Flow Chart: Example of Initial Settings of a PMGI

39.3.2.2 PHY Register Reading Operation

Figure 39.5 is a flow chart giving an example of how to read a register and Figure 39.6 shows an example of signals in register reading.

The setting procedure for register reading is as follows.

- (1) Initialize the PMGI. Refer to section 39.3.2.1, Initializing a PMGI.
- (2) Set the PSMR.PSMAD bit to 0 to select reading.
- (3) Set the PHY address in the PSMR.PDA[4:0] bits.
- (4) Set the PHY register address in the PSMR.PRA[4:0] bits.
- (5) Set the PSMR.PSME bit to 1 to start transfer of the management frame. After transfer is completed, the setting of the PSMR.PSME bit automatically becomes 0 and an access completed interrupt request is generated.
- (6) Confirm that the PSMR.PSME bit is 0 and then read the PSMR.PRD[15:0] bits.

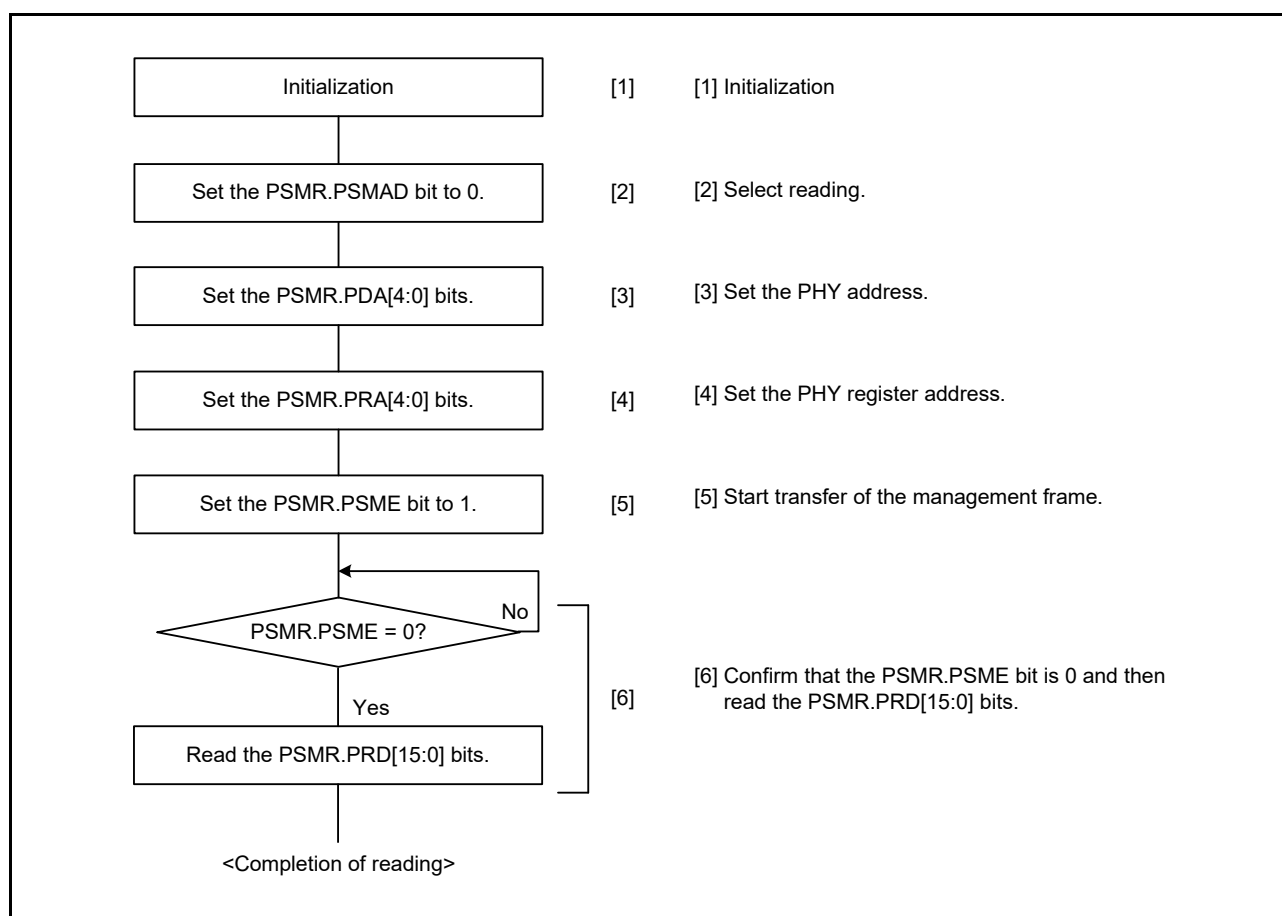


Figure 39.5 Flow Chart: Example of PHY Register Reading

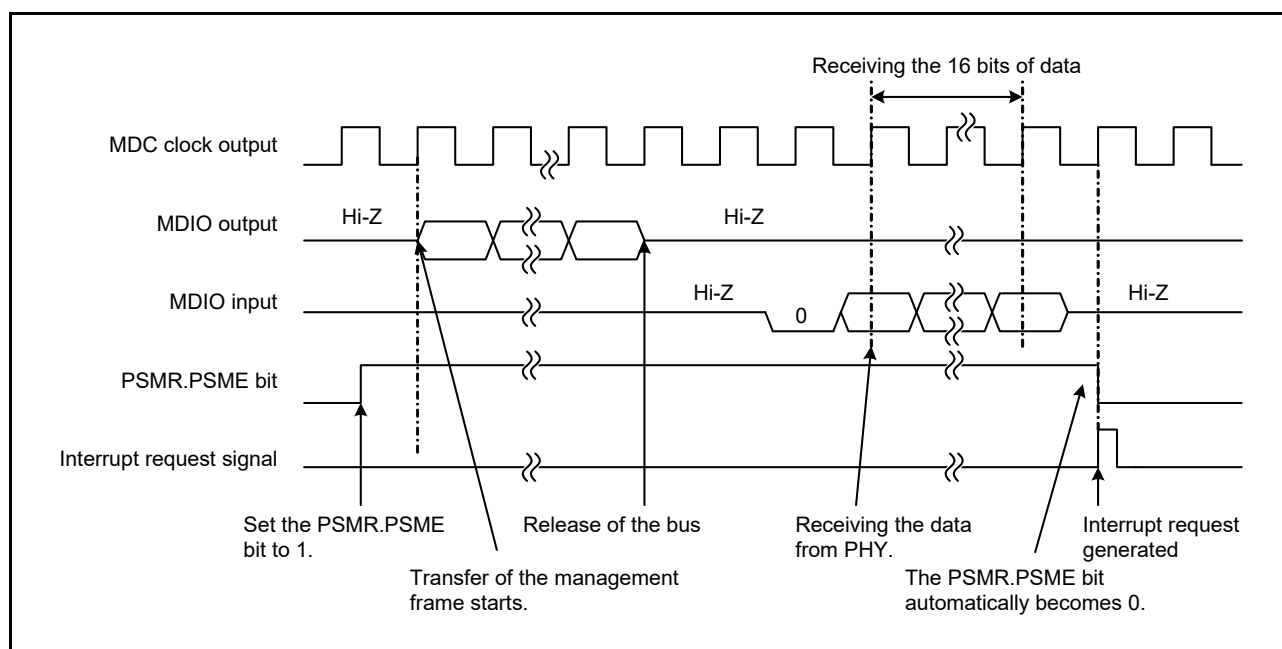


Figure 39.6 Example of PHY Register Reading Operation

39.3.2.3 PHY Register Writing

Figure 39.7 is a flow chart giving an example of how to write to a register and Figure 39.8 shows an example of signals in register writing.

The setting procedure for register writing is as follows.

- (1) Initialize the PMGI. Refer to section 39.3.2.1, Initializing a PMGI.
- (2) Set the PSMR.PSMAD bit to 1 to select writing.
- (3) Set the PHY address in the PSMR.PDA[4:0] bits.
- (4) Set the PHY register address in the PSMR.PRA[4:0] bits.
- (5) Set the value to be written to the PHY register in the PSMR.PRD[15:0] bits.
- (6) Set the PSMR.PSME bit to 1 to start transfer of the management frame.

After transfer is completed, the setting of the PSMR.PSME bit automatically becomes 0 and an access completed interrupt request is generated.

- (7) Confirm that the PSMR.PSME bit is 0.

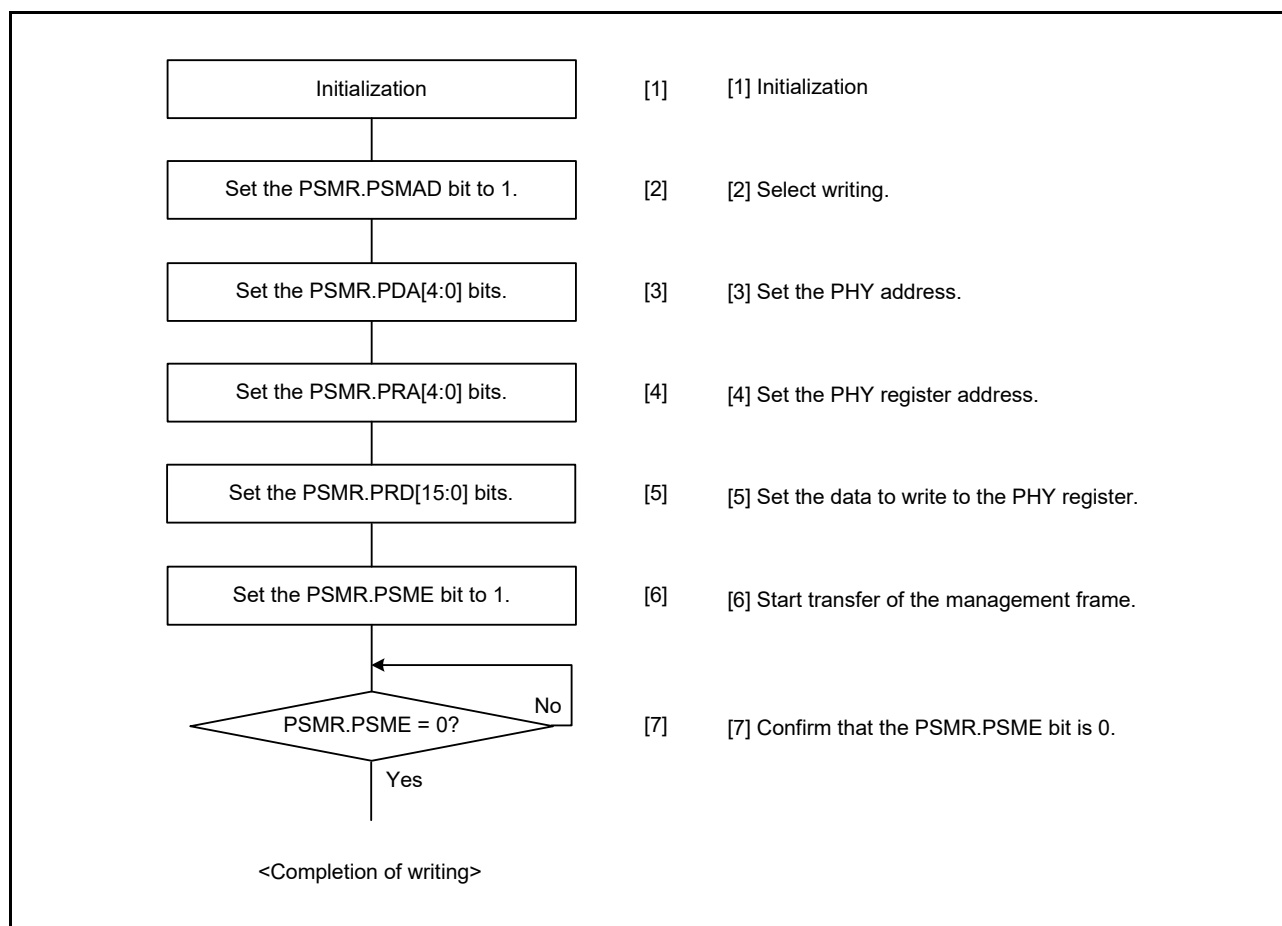


Figure 39.7 Flow Chart: Example of PHY Register Writing

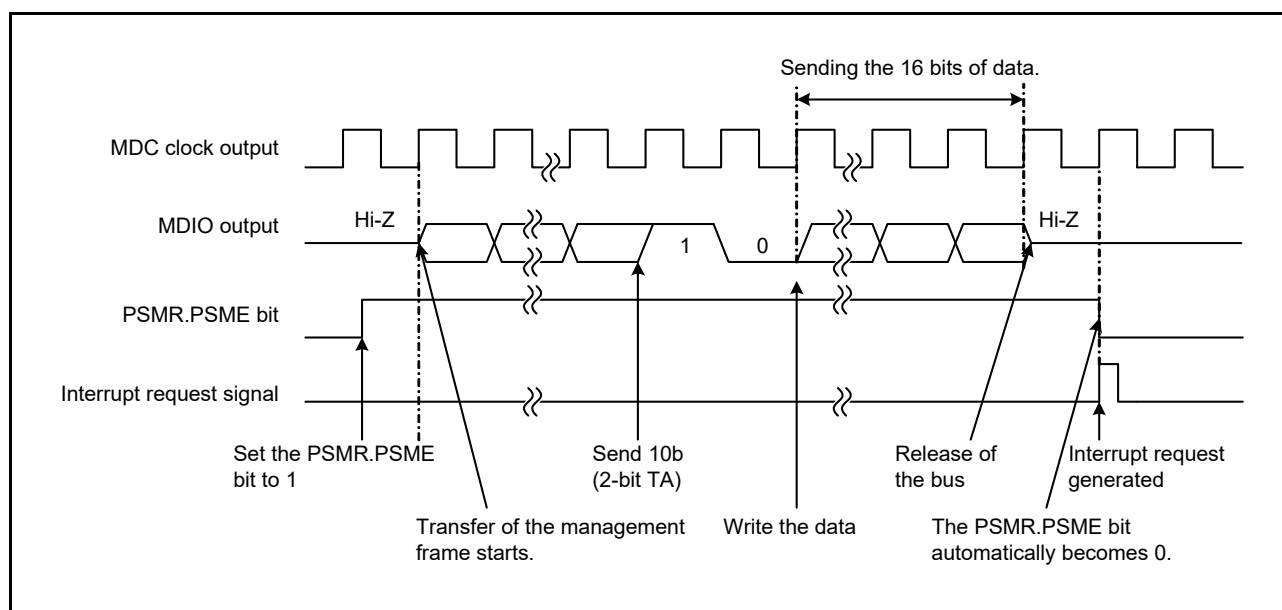


Figure 39.8 Example of PHY Register Writing Operation

39.3.3 Setting the Management Data Clock (MDC)

The PCLKA clock is frequency-divided to generate the MDC. Select the division ratio in the PMGCR.PSMCS[5:0] bits. Obtain the frequency of the MDC by calculating $PCLKA \text{ (MHz)} / ((PMGCR.PSMCS[5:0] + 1) \times 2)$. Table 39.4 lists examples of settings for the MDC frequency.

Table 39.4 Examples of Settings for the MDC Frequency

PCLKA frequency	PMGCR.PSMCS[5:0]	MDC frequency
25 MHz	04h	2.5 MHz
50 MHz	09h	2.5 MHz
80 MHz	0Fh	2.5 MHz
96 MHz	13h	2.4 MHz
100 MHz	13h	2.5 MHz
120 MHz	17h	2.5 MHz

39.3.4 Adjusting the Timing of Management Data

The timing of output on the MDIO can be delayed and the timing for capturing the MDIO input can be adjusted. Set appropriate values in consideration of the electrical characteristics of the PHY LSI chip and the propagation delay on the board.

(1) Adjusting the hold time for the management data

Figure 39.9 shows the timing when the hold time is adjusted with a delay of 2 PCLKA cycles.

Outputting the MDIO 2 PCLKA cycles after the rising edge of the MDC is selectable by setting the PMGCR.PSMHT[2:0] bits to 010b.

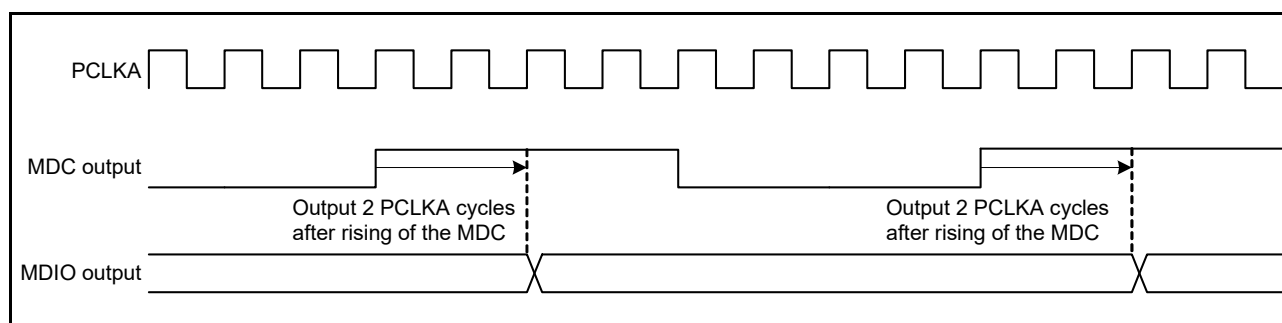


Figure 39.9 Timing with the Hold Time Adjusted for a Delay of 2 PCLKA Cycles

(2) Adjusting the timing of the capture of management data

Figure 39.10 shows the capture timing when the capture of the MDIO input is to proceed 2 PCLKA cycles before the MDC rising.

Capturing the MDIO input 2 PCLKA cycles earlier than the rising edge of the MDC is selectable by setting the PMGCR.PSMCT[2:0] bits to 010b.

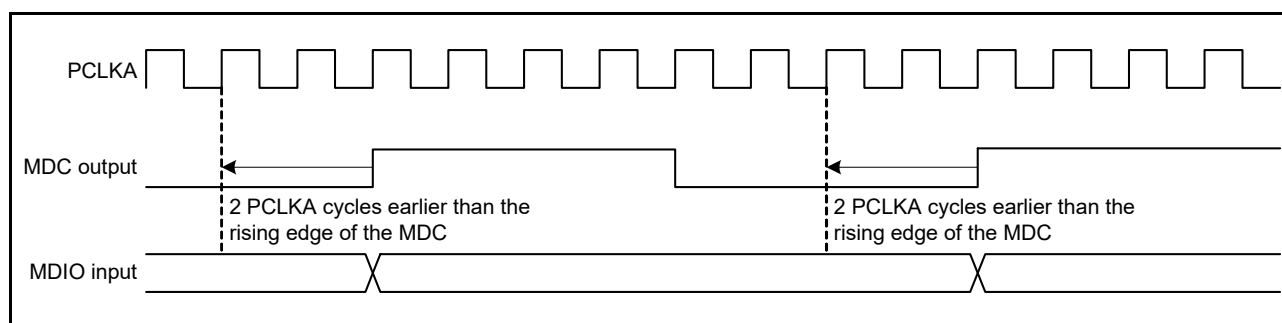


Figure 39.10 Capturing Timing that is to be 2 PCLKA Cycles Earlier

39.3.5 Interrupts

When transfer of an MII management frame is completed, the PSMR.PSME bit automatically becomes 0, and an access-completed interrupt request is generated.

Each PMGI (PMGI0I and PMGI1I) has its own access completed interrupt circuit, and different vector addresses are allocated to the two. When an access completed interrupt is generated, the corresponding interrupt request is output.

When the interrupt request is to generate a CPU interrupt, the order of priority of the two signals can be changed by setting the interrupt controller. For details, refer to [section 15, Interrupt Controller \(ICUD\)](#).

Table 39.5 **Interrupt Sources of PMGI**

Name	Interrupt Source	Starting the DTC	Starting the DMAC
PMGI0I	Access by PMGI0 completed	Not possible	Not possible
PMGI1I	Access by PMGI1 completed	Not possible	Not possible

39.4 Usage Notes

39.4.1 Settings of the Module Stop Function

Each PMGI can be placed in or released from the module stop state by setting module stop control register B (MSTPCRB). The initial setting is for the module stop state. Release from the module stop state enables access to the registers of the PMGI. For details of module stop control register B, refer to **section 11, Low Power Consumption**.

39.4.2 Note on setting PSMHT

Do not set the PMGCR.PSMHT[2:0] bits to a value greater than that of the PSMCS[5:0] bits.

39.4.3 Note on setting PSMCT

Do not set the PMGCR.PSMCT[2:0] bits to a value greater than that of the PSMCS[5:0] bits.